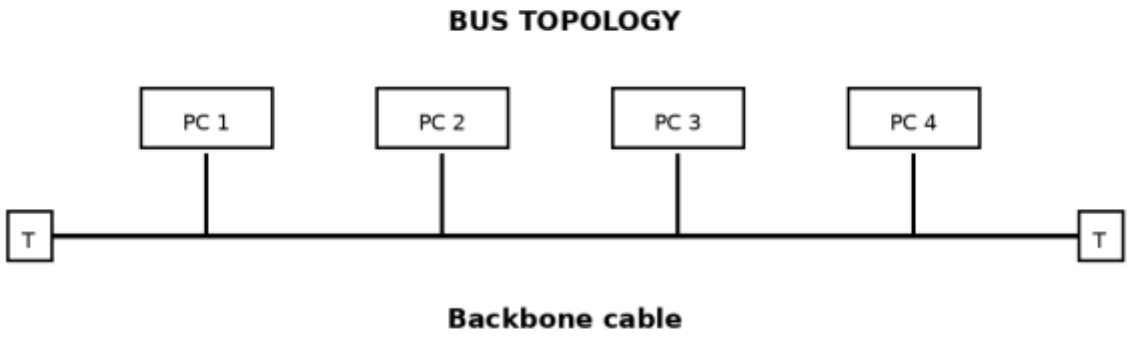
DAY 2 NETWORK TOPOLOGIES THEORY

Task 1: Draw a diagram of a bus topology and label all key components, then write one real-world example where this topology might be used.

Definition

Bus topology is a network arrangement in which all computers and devices are connected to one main cable called the backbone. Terminators are placed at both ends of the cable to prevent signal reflection.

Diagram



Key Components

- Computers or nodes
- Backbone cable
- Connectors or cable taps
- Terminators at both ends

Real-World Example

Bus topology was used in older Ethernet networks, such as 10BASE2 and 10BASE5, where several computers shared a single coaxial cable.

Conclusion

In bus topology, all devices share one main cable. It is simple and uses less cable, but a fault in the backbone can affect the whole network.

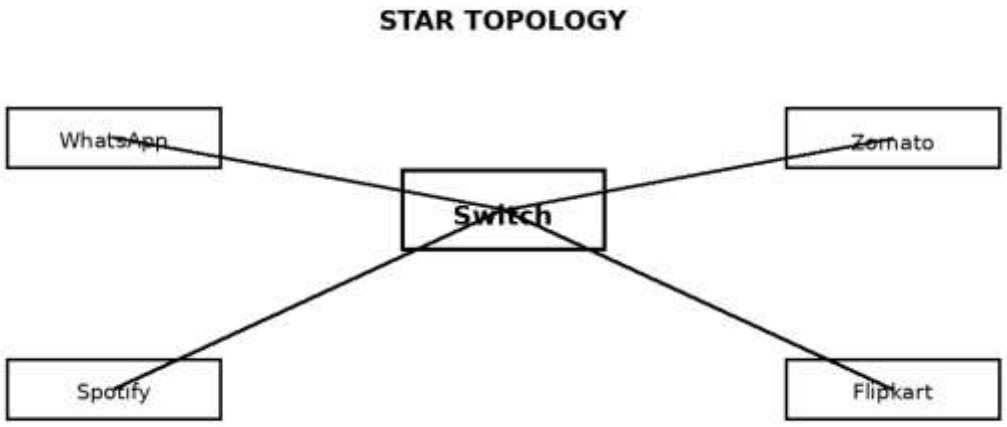
DAY 2 NETWORK TOPOLOGIES THEORY

Task 2: In Cisco Packet Tracer, create a simple star topology with one switch and four PCs connected to it; name each PC after a popular app (e.g., WhatsApp, Zomato, Spotify, Flipkart) and take a screenshot of your topology.

Aim

To create a star topology using one switch and four PCs in Cisco Packet Tracer.

Diagram / Expected Layout



Procedure

1. Open Cisco Packet Tracer.
2. Place one 2960 switch in the workspace.
3. Place four PCs in the workspace.
4. Rename the PCs as WhatsApp, Zomato, Spotify, and Flipkart.
5. Choose Copper Straight-Through from the Connections menu.
6. Connect each PC to a separate port on the switch.
7. Wait for the link indicators to turn green. If necessary, configure IP addresses in the same network.
8. Take a screenshot of the completed topology and attach it to this assignment.

Result

A star topology with one central switch and four PCs was created in Cisco Packet Tracer. Each PC connects directly to the switch.

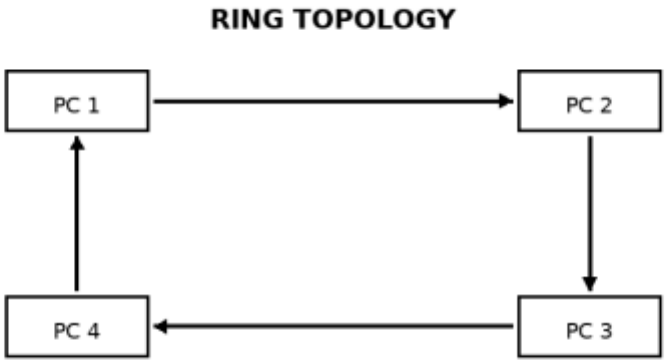
DAY 2 NETWORK TOPOLOGIES THEORY

Task 3: Draw a ring topology diagram and explain in 2-3 sentences how data travels between devices in this setup.

Definition

Ring topology is a network arrangement in which each device is connected to two neighbouring devices, forming a closed loop.

Diagram



How Data Travels

Data moves from one device to the next around the ring in the direction indicated by the arrows. Each device forwards the data until it reaches the intended destination. In some ring networks, data may travel in the opposite direction or use a second ring for resilience.

Example

Older Token Ring networks and some industrial communication networks have used ring-based arrangements.

Conclusion

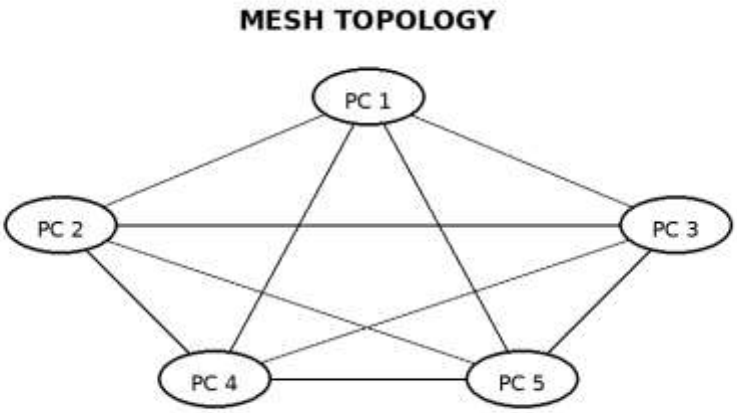
A ring topology forms a closed loop. Its performance and reliability depend on the ring design and the mechanisms used to handle a failed link or device.

DAY 2 NETWORK TOPOLOGIES THEORY

Task 4: Compare mesh and star topologies by listing 2 advantages and 2 disadvantages of each in a table format.Mesh topology connects devices through multiple links, while star topology connects each device to a central switch or hub.

Topology	Advantages	Disadvantages
Mesh	1. Highly reliable because multiple paths may be available. 2. Failure of one link may not stop communication.	1. Expensive because it needs many cables and ports. 2. Installation and maintenance are complex.
Star	1. Easy to install, manage, and troubleshoot. 2. Failure of one PC cable usually does not affect other PCs.	1. If the central switch fails, the network is disrupted. 2. Requires more cable than a bus topology.

Mesh Topology Diagram (Reference)



Overall Conclusion

Bus, star, ring, and mesh are common network topologies. The best topology depends on cost, reliability, installation complexity, and the requirements of the network.